

Alex's Quest

Cross-Platform Mobile App

PROJECT TYPE Cross-Platform Mobile App	PLATFORM Cross-platform mobile application
AUDIENCE Grade-school English learners and teachers	PRIMARY FOCUS Grammar, sentence construction, and pronunciation

Overview

A story-driven English learning mobile app that guides grade-school learners through five chapters of an adventurous city-to-jungle quest with a parrot named Alex. Learners practice sentence construction, grammar, and pronunciation with real-time on-device speech scoring. Built cross-platform with Expo SDK 55, React Native 0.83, and a bundled whisper.rn model so student audio never leaves the device.

Core Features

- 5 story-based chapters mapped to grammar learning objectives (sentence construction to full application)
- On-device Whisper transcription (whisper.rn) - student audio never leaves the device
- Three-factor scoring (Content 50% / Pronunciation 40% / Fluency 10%) with phoneme-level feedback via CMU dictionary + Needleman-Wunsch alignment
- Multi-question types: Identify, Build, Speak, Choice, Rival - with adaptive escalating hints
- Story map with animal companion rewards, badges, and certificate of completion
- PIN-gated Teacher Portal with assessment history, transcripts, and per-student analytics
- Lottie character animations and chapter-specific background music
- Zustand-backed local progress, expo-router typed navigation, EAS Build pipeline

Experience and Design

- Five story chapters move from sentence construction to complete grammar application.
- Identify, Build, Speak, Choice, and Rival questions provide varied assessment formats.
- Content, pronunciation, and fluency scoring produce detailed, tiered feedback.
- Story maps, animal companions, badges, and certificates reward sustained progress.

Architecture and Data

- Expo and React Native provide the cross-platform interface and typed navigation.
- Zustand stores learner progress and teacher records locally on the device.
- Whisper runs on-device for speech transcription and pronunciation assessment.
- Reanimated, Lottie, audio, and video modules deliver story and character presentation.

Security and Privacy

- Student speech is processed on-device and is not uploaded to a remote service.
- Local progress avoids unnecessary account, telemetry, and cloud data collection.
- A teacher PIN restricts access to assessment history and learner analytics.
- Microphone access is active only during answer capture and audio buffers are cleaned afterward.

Technology Stack

Expo SDK 55	Application capability
React Native 0.83	Application framework and UI
React 19	Application framework and UI
TypeScript	Application language
NativeWind	Interface styling and design system
Zustand	State, data flow, and architecture
whisper.rn	AI and speech processing
Reanimated + Lottie	Animation and interaction

Performance and Reliability

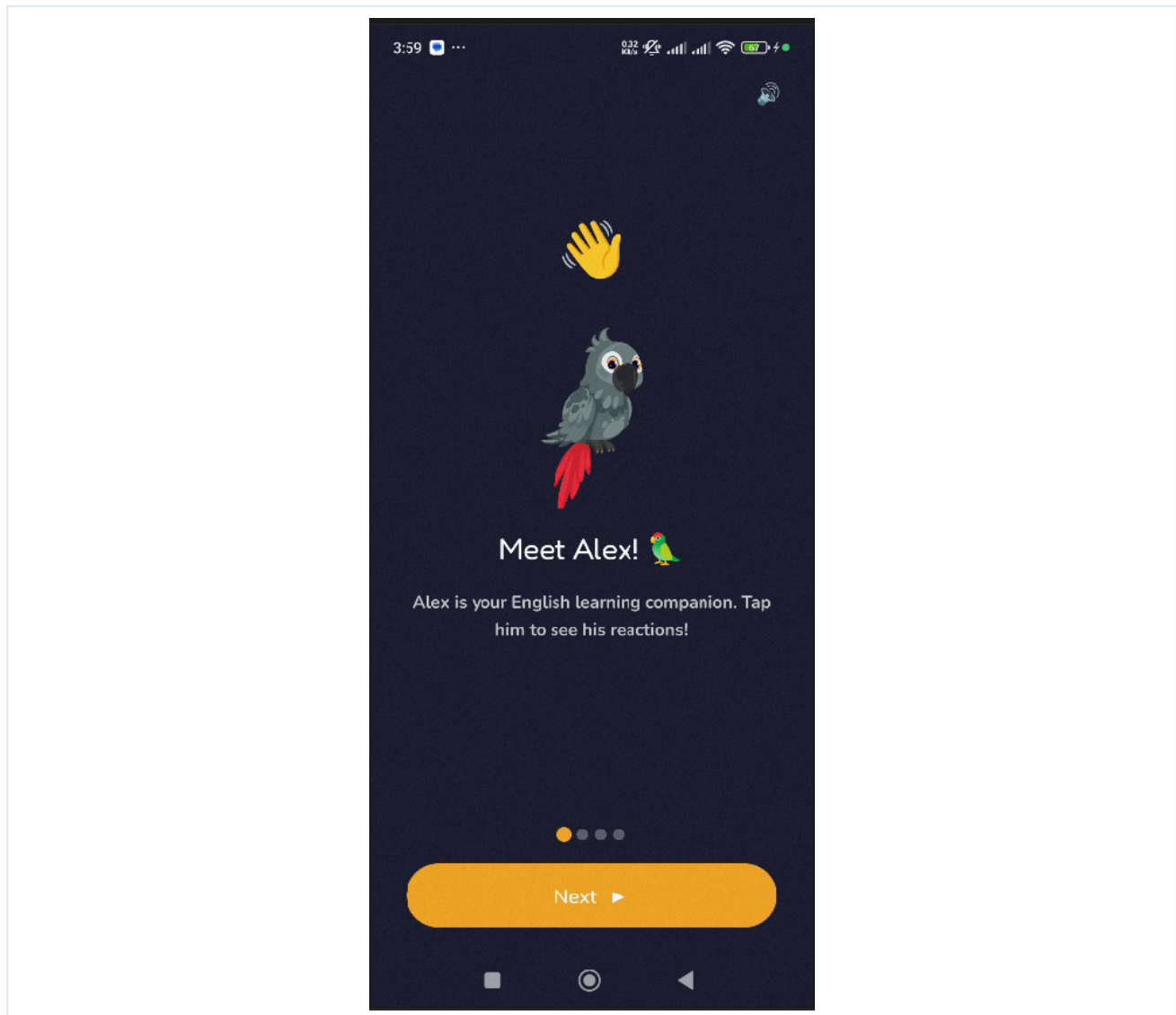
- The native Whisper module avoids slower JavaScript-only speech processing.
- Chapter media loads only when the related story section is opened.
- Persistent local state allows learners to resume without repeating completed work.
- Audio and animation resources are released when screens unmount.

Platform Requirements

- Supported Android or iOS device with enough storage for the bundled speech model.
- Microphone permission for Speak questions and pronunciation scoring.
- Audio output for narration, vocabulary playback, and chapter music.
- Core learning and speech assessment can operate without uploading student audio.

PAGE OVERVIEW

Onboarding

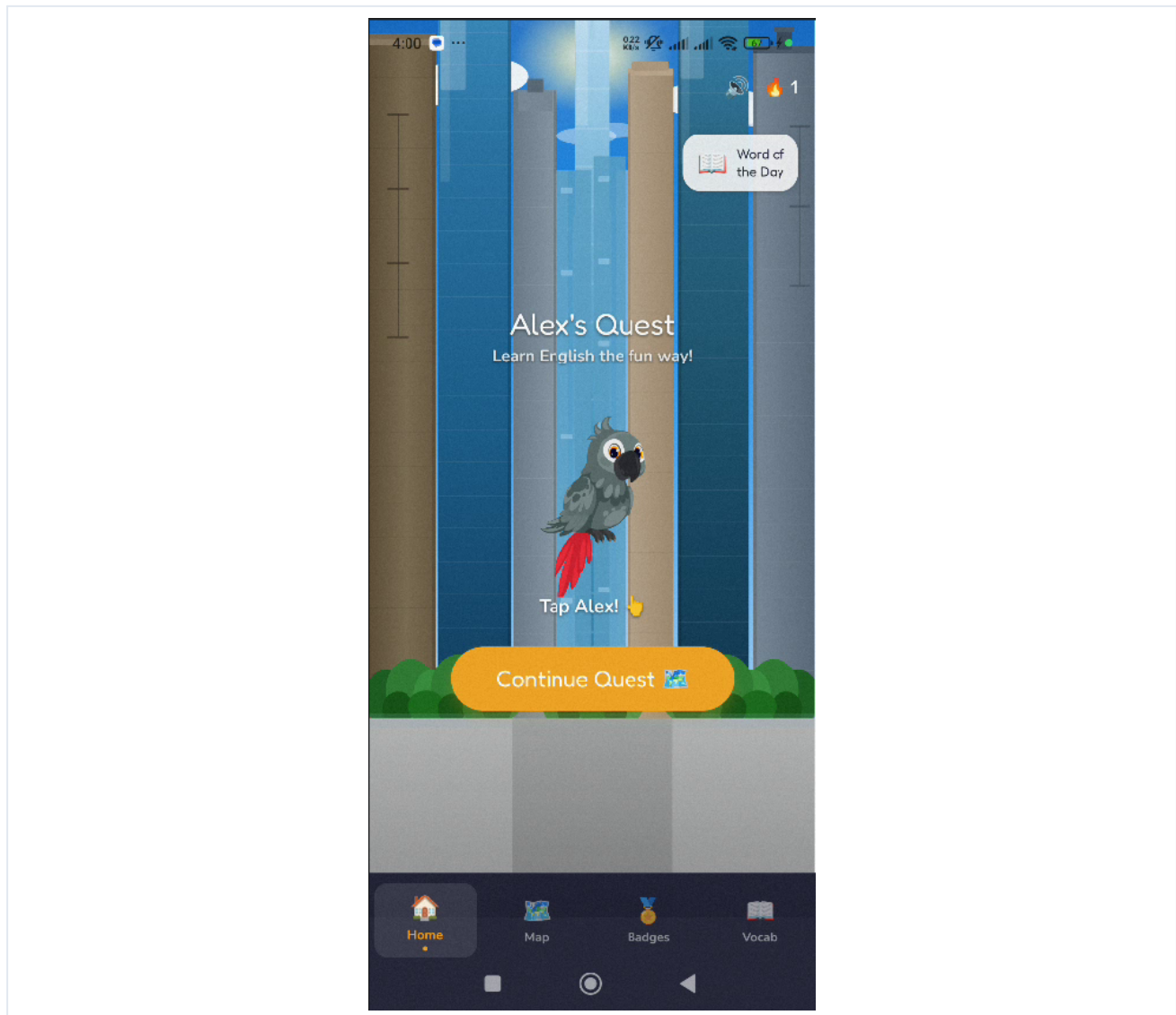


Onboarding

Interface view from Alex's Quest.

PAGE OVERVIEW

Home

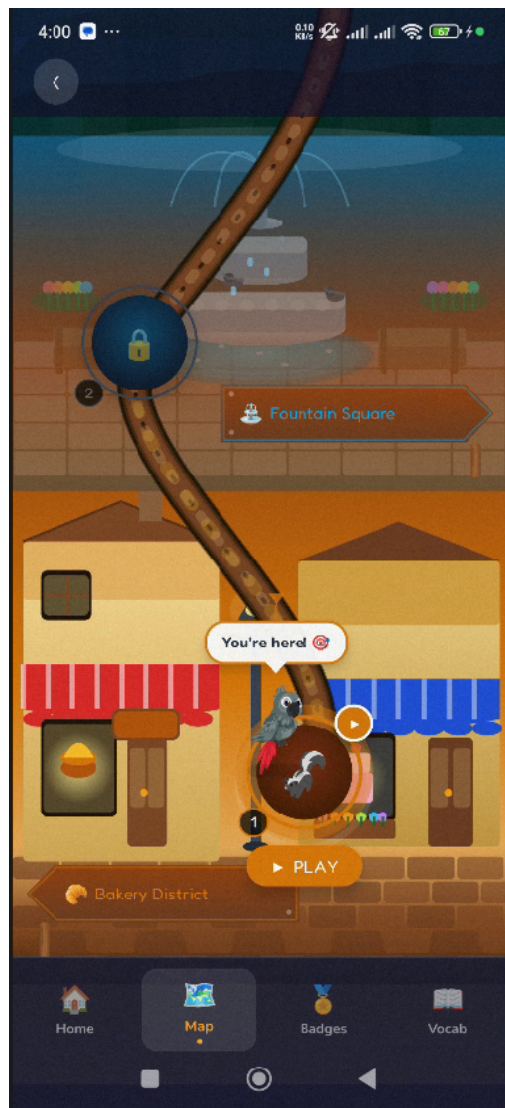


Home

Interface view from Alex's Quest.

PAGE OVERVIEW

Story Map

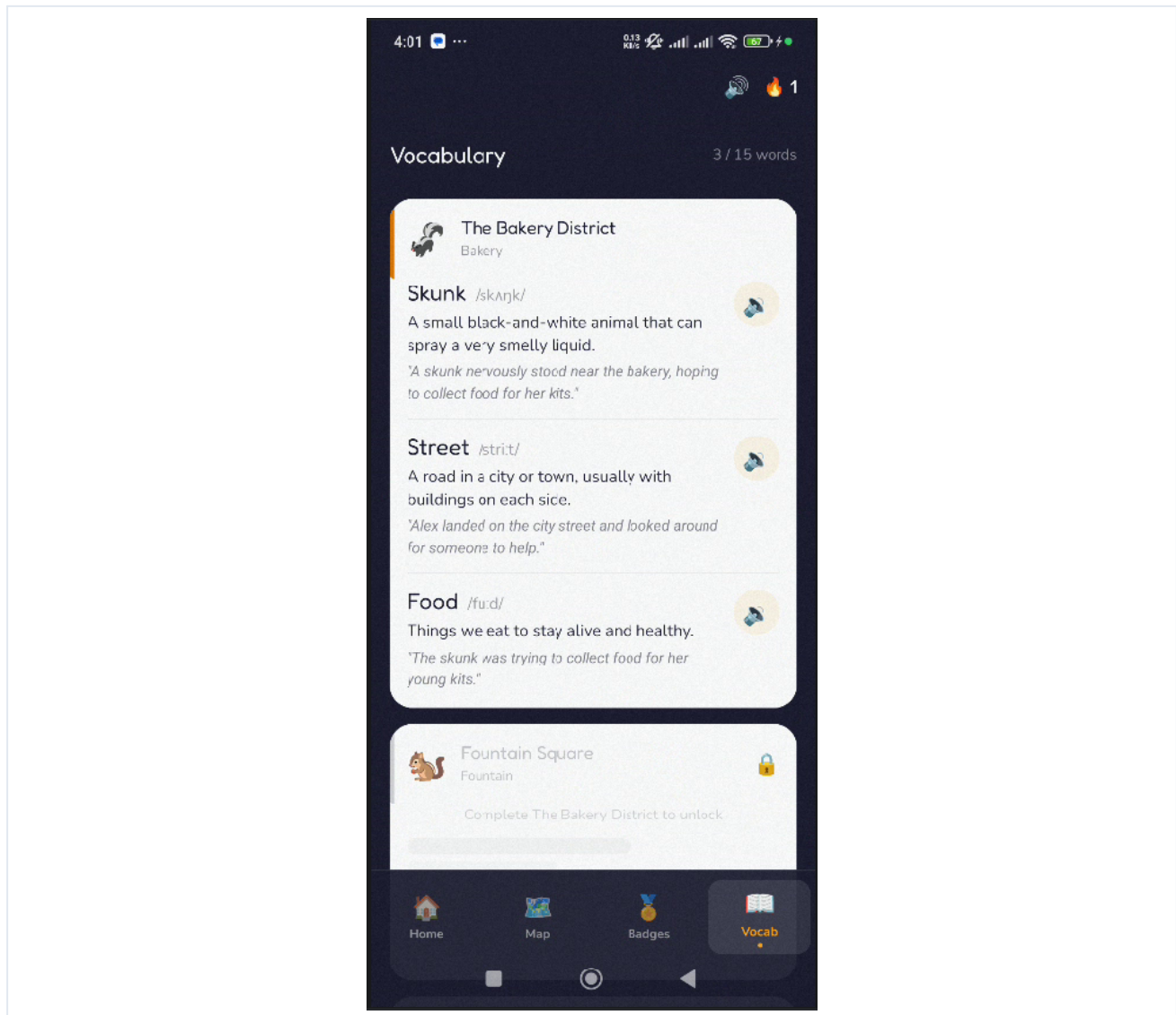


Story Map

Interface view from Alex's Quest.

PAGE OVERVIEW

Vocabulary

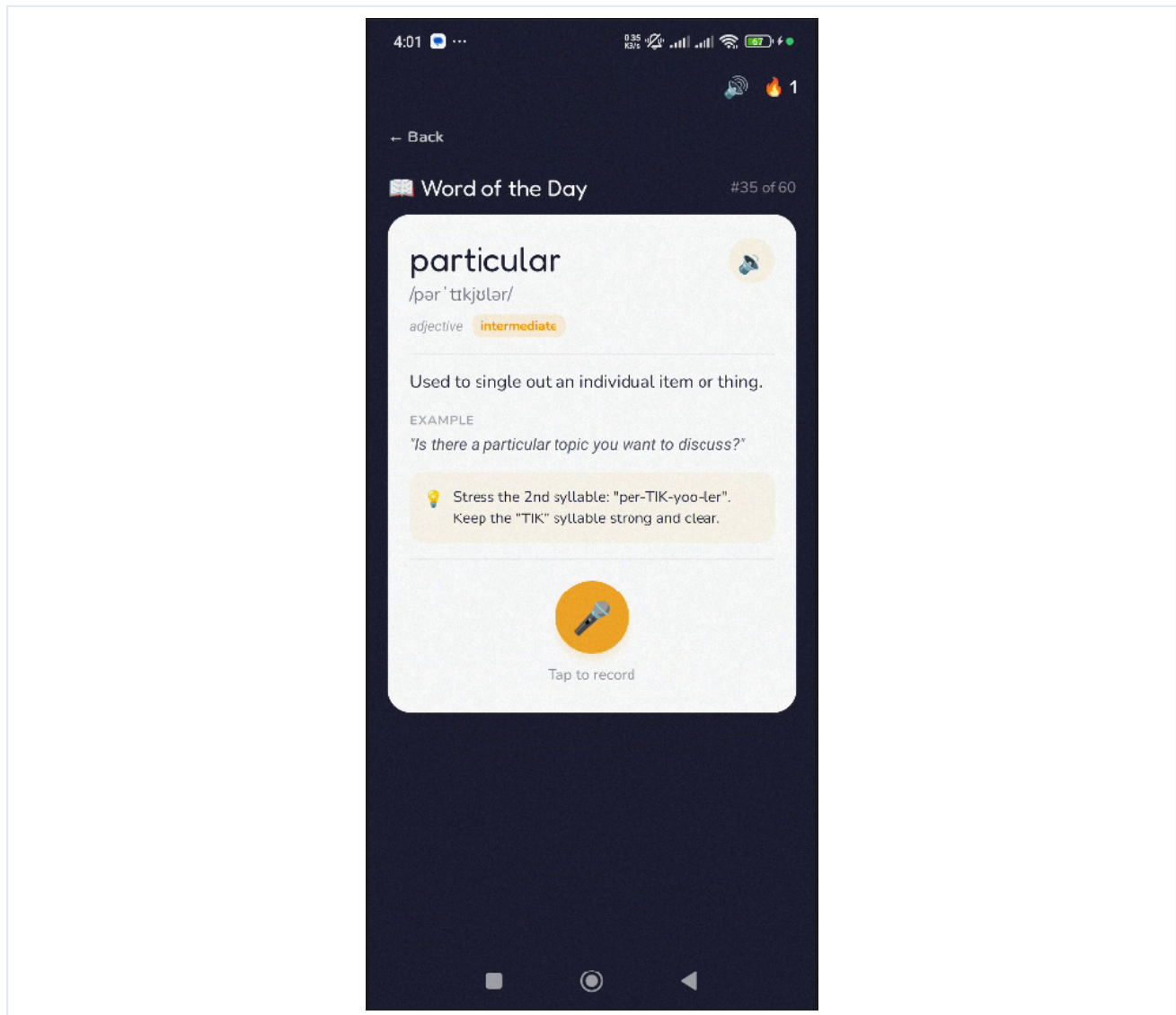


Vocabulary

Interface view from Alex's Quest.

PAGE OVERVIEW

Word of the Day

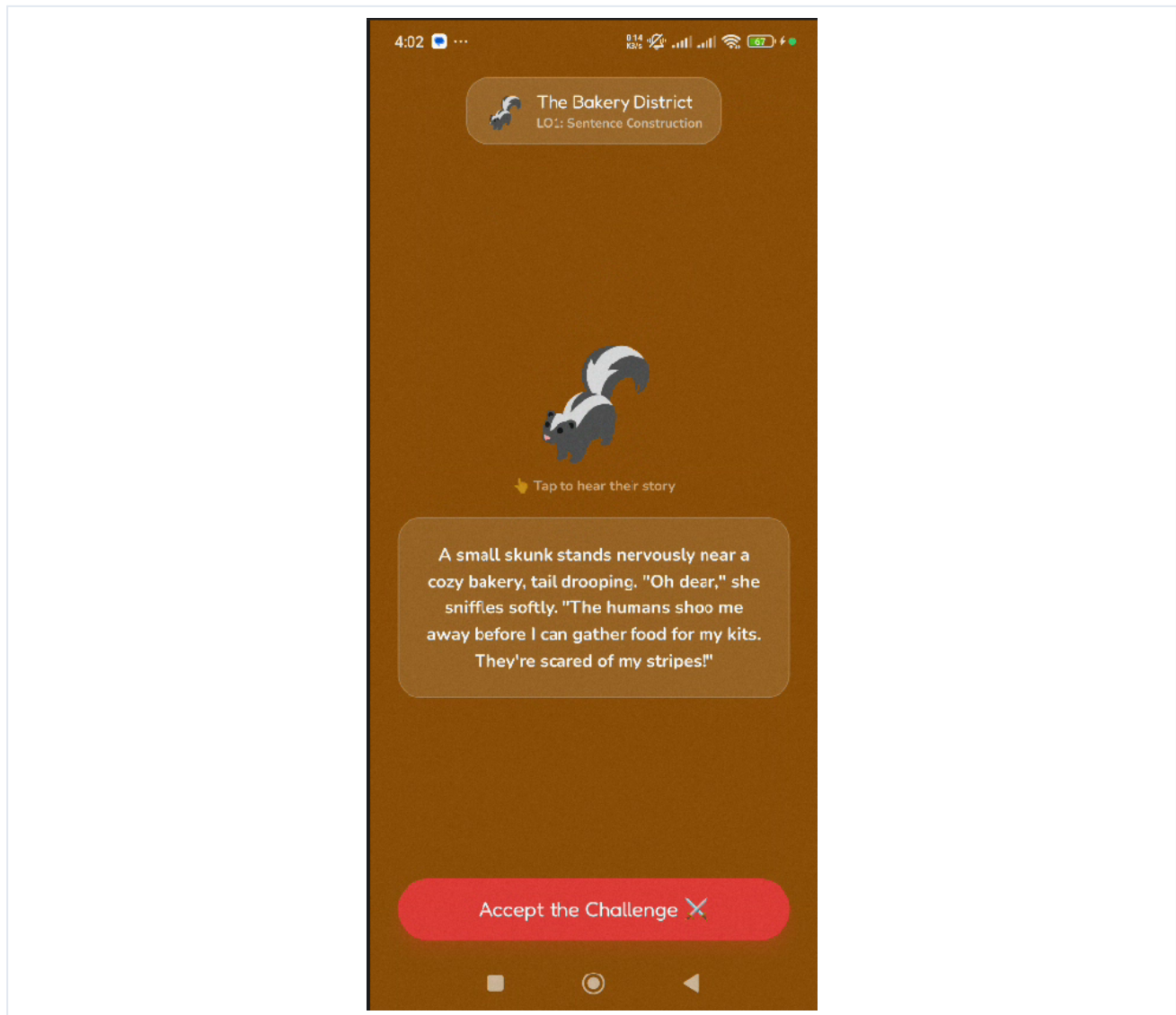


Word of the Day

Interface view from Alex's Quest.

PAGE OVERVIEW

Chapter Challenge

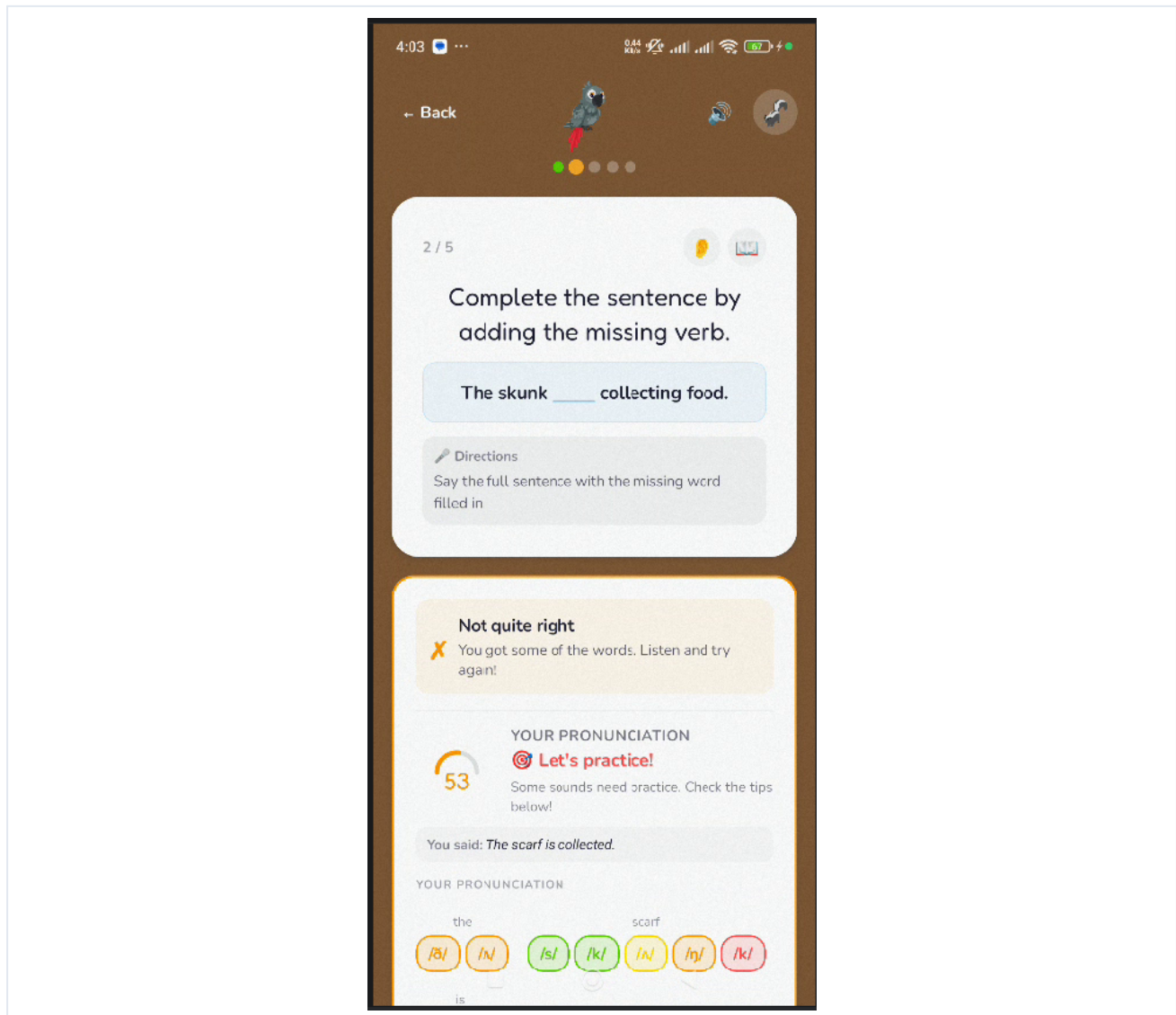


Chapter Challenge

Interface view from Alex's Quest.

PAGE OVERVIEW

Assessment - Pronunciation Feedback

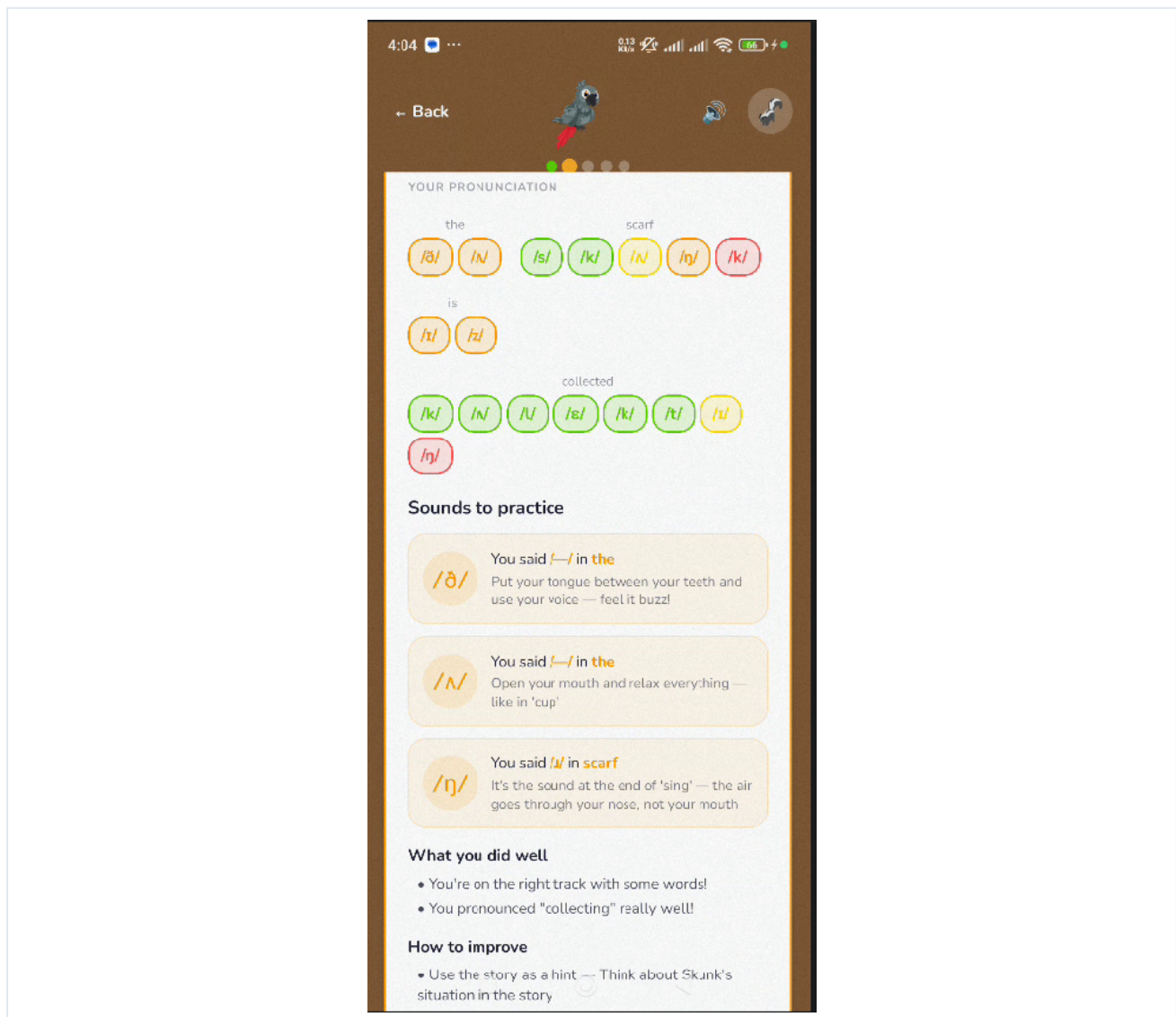


Assessment - Pronunciation Feedback

Interface view from Alex's Quest.

PAGE OVERVIEW

Assessment (cont.)

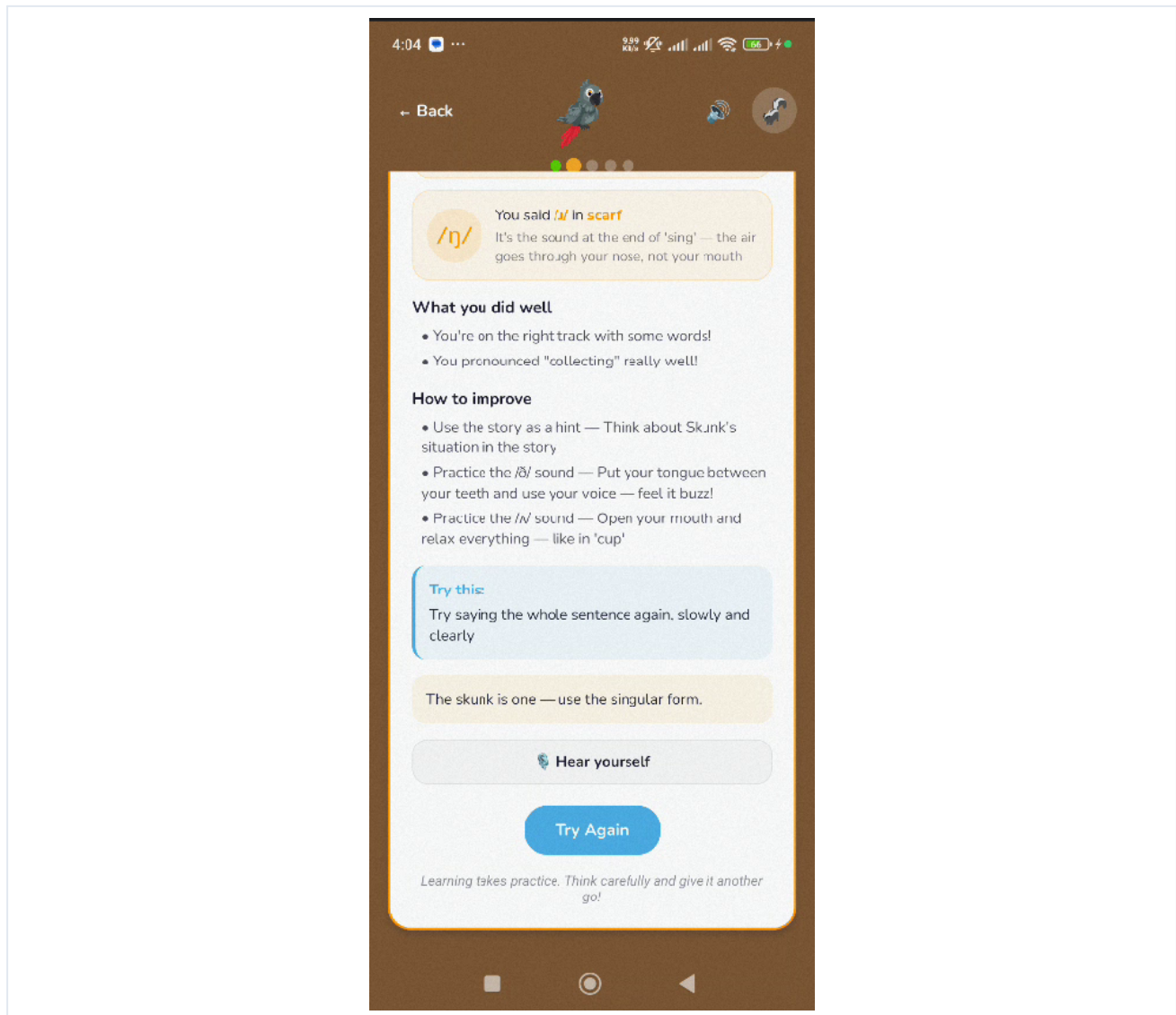


Assessment (cont.)

Interface view from Alex's Quest.

PAGE OVERVIEW

Assessment (cont.)

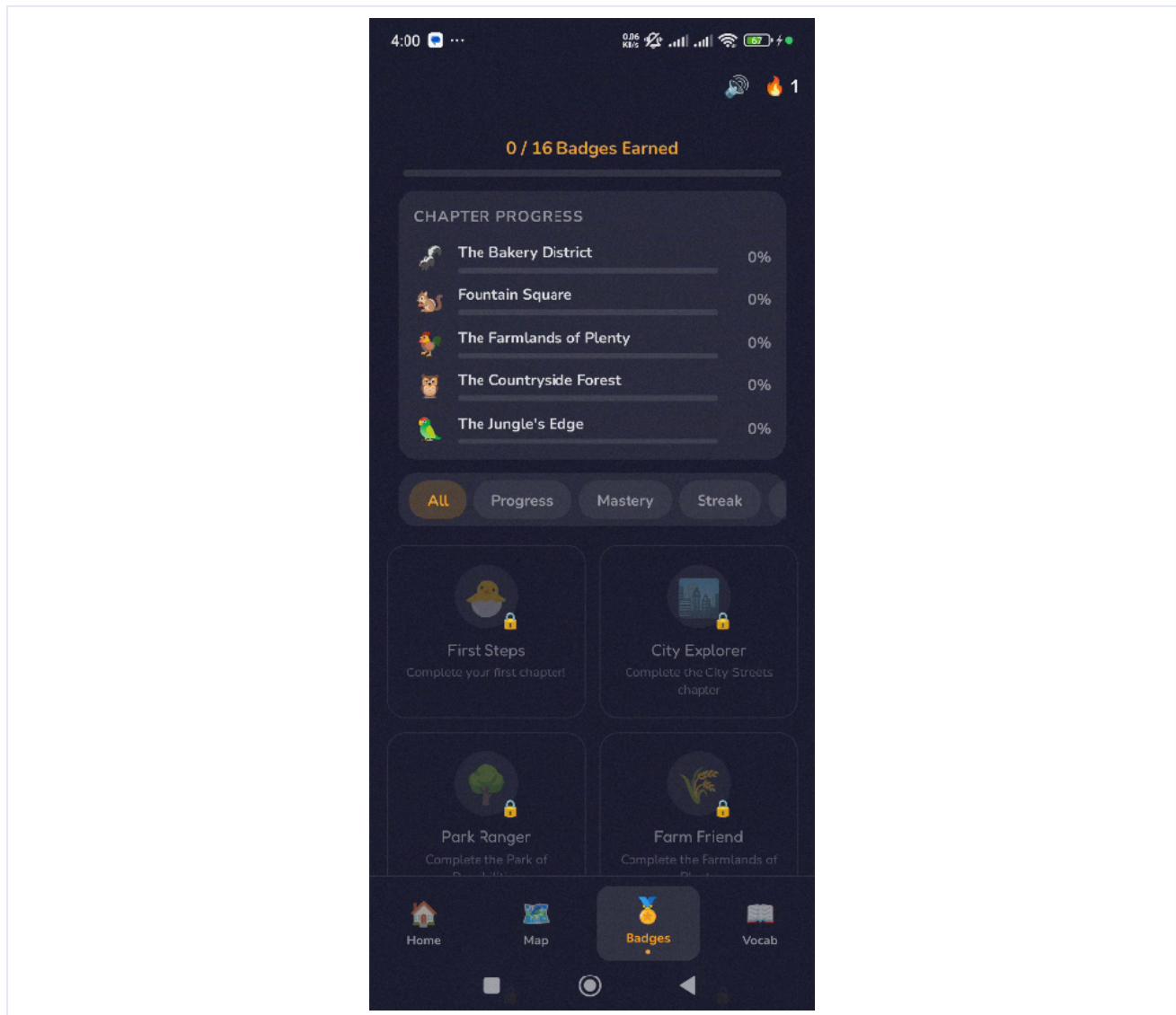


Assessment (cont.)

Interface view from Alex's Quest.

PAGE OVERVIEW

Badges & Progress



Badges & Progress

Interface view from Alex's Quest.